

부록 — 프롬프트 전문 · 실험 데이터 · Few-shot 악화 케이스 분석

SOAP 노트 검증 1조 · 2026-07-22 · 본편 PDF(프롬프트개선_결과공유_0722)의 상세 부록

1. 실험별로 쓴 데이터와 프롬프트 매핑

실험 (그래프)	데이터	프롬프트/파이프라인
단계 사다리 (fig1 · fig1b · fig1c) 노이즈 3선 (fig2)	ACI-Bench valid 20건 (공식 벤치마크, gold=전문 작성 노트) × 4모델 김고은 형태노이즈 100건 (자체 제작, gold=고은 작성): 제3화자 24 · ASR 화자 오태깅 24 · 시간순서 18 · 재수량화 18 · ASR수치 16 (ID 접두어 N_G/N_M/N_T/N_R/N_N) × Qwen 7B/32B	baseline → 개선1 → 개선2 → 개선3 (=개선2 출력에 32B 검증기 자동교정; 프롬프트가 아닌 추론 파이프라인) soap(기존 ACI 프롬프트) → soap_hard(+방어 규칙 7종) → soap_fewshot(+실패모드 예시 3종)
참고: 의미노이즈	장다연 의미노이즈 clean/noisy 64쌍 (flat128)	전 단계 평가했으나 모든 전략이 거의 완벽 방어(전략 간 차 <1%p)라 그래프 제외
참고: few-shot v2	김고은 few-shot 예시 18종 — 평가 데이터가 아니라 프롬프트 재료 (평가셋과 무중복 명시)	soap_fewshot_v2(18예시 전체 삽입) — 양 모델 악화로 미채택, 본편 PDF §3 참조

환각률 판정은 모든 실험 공통: Qwen2.5-32B LLM Judge가 문장 단위로 “대화에 근거 있는가”를 판정.

2. 두 그래프의 프롬프트는 동일한가?

동일하지 않고, 계보(상속) 관계다. 단계 그래프는 baseline → 개선1 → 개선2(+교정)를, 노이즈 그래프의 세 선은 모두 soap 기반을 쓴다. soap은 개선2를 계승한 발표 최종 규격이다:

- 공유하는 것(개선2 = soap): 턴 번호 입력 ([T0], [T1], ...) · 문장별 [T#] 인용 강제 · 발화행위 기준 라우팅 · 확실성 보존 · [미확인] 규칙 — 검증에 관련된 장치는 전부 동일.
- 다른 것: 출력 형식만. 개선2는 6개 평면 헤더 (CHIEF COMPLAINT ... ASSESSMENT AND PLAN), soap은 명시적 S:/O:/A:/P: 배너 + 세부헤더 (EMR 등재용 최종 규격).

노이즈 실험이 soap을 베이스로 한 이유: soap_hard · soap_fewshot이 soap에 규칙/예시만 얹은 통제 A/B이기 때문 (같은 베이스 위 비교라야 “규칙 vs 예시” 효과가 분리됨). 완전히 동일하게 맞추려면 개선2 위에 +규칙/+예시 변형을 새로 만들어 노이즈 100건을 재평가해야 한다 (모델 2종 기준 GPU 약 1 - 2시간). 다만 검증 장치가 동일하므로 현재 구성으로도 “개선2 계열 최종 프롬프트의 노이즈 강건성 실험”으로 일관되게 설명 가능하다.

3. 단계 사다리 프롬프트 전문

3.1 Baseline (prompts/baseline.txt) — 기본 지시만

Summarize the following doctor-patient conversation into a clinical note.

3.2 개선 1 · 프롬프트 엔지니어링 (improve1_prompt.txt) — 규칙 명문화

You are a clinical scribe. Convert the doctor-patient conversation into a structured SOAP note.

Use these UPPERCASE section headers (omit a header only if truly nothing applies):

CHIEF COMPLAINT
HISTORY OF PRESENT ILLNESS
REVIEW OF SYSTEMS
PHYSICAL EXAMINATION
RESULTS
ASSESSMENT AND PLAN

Rules:

- Write ONLY what is supported by the conversation. Do NOT invent findings, vital signs, lab values, diagnoses, medications, or plans that were not stated.
- Do not add boilerplate normal findings that were never discussed.
- Keep it concise and clinical.

Output the note only.

3.3 개선 2 · 컨텍스트 엔지니어링 (improve2_context.txt) — 턴 번호·문장별 인용

You are a clinical scribe producing a verifiable SOAP note. Every statement must be traceable to the conversation.

The transcript turns are numbered [T0], [T1], ... Use these numbers as evidence.

SECTION FORMAT — use these UPPERCASE headers:

CHIEF COMPLAINT
HISTORY OF PRESENT ILLNESS
REVIEW OF SYSTEMS
PHYSICAL EXAMINATION
RESULTS
ASSESSMENT AND PLAN

RULE 1 — Classify by speech act, not by speaker.

Whoever is speaking, route content by what the utterance DOES:

- symptom report / history / social history -> SUBJECTIVE (CHIEF COMPLAINT, HISTORY OF PRESENT ILLNESS, REVIEW OF SYSTEMS)
 - exam finding, vital sign, measured value -> OBJECTIVE (PHYSICAL EXAMINATION, RESULTS)
 - diagnosis / differential / severity -> ASSESSMENT
 - order / prescription / follow-up / education -> PLAN
- (A doctor dictating the patient's history is still SUBJECTIVE.)

RULE 2 — Preserve certainty. Do not upgrade uncertainty to fact.

Keep the original assertion level using clinical hedges (i2b2 assertion classes):

- present -> state it plainly
 - absent (patient denies) -> "denies ...", "no ..."
 - possible (maybe/likely/suspected) -> "reports possible ...", "likely ...", "suspected ..."
 - conditional / hypothetical -> keep the condition ("if ... then ...")
- If the patient was unsure ("I'm not sure", "maybe"), do NOT write it as definite.

RULE 3 — Mark the unknown; never fill it.

If a section's information is not present in the conversation, write exactly:

[미확인: not mentioned in conversation]

Do NOT fabricate plausible-but-unstated content to complete a section.

RULE 4 — Cite evidence.

End every sentence with the supporting turn id(s) in brackets, e.g. "... hypertension. [T3]" or "[T3, T5]".

Only cite turns that actually contain the information.

Output the note only.

3.4 개선 3 · 검증기 자동교정 — 프롬프트가 아닌 파이프라인

개선 2의 출력을 32B 검증기가 문장별 판정 → 근거 없음 (RED) 문장을 원 생성모델이 아래 지시로 재작성 → 재판정 통과 시에만 유지(실패 시 삭제). 수술로 비는 구역엔 [미확인] 삽입.

You fix ONE sentence of a clinical SOAP note so that it is fully supported by the conversation transcript. Return ONLY the corrected sentence, ending with the supporting turn citation(s) like [T3] or [T3, T5]. Keep clinical register. Do NOT add information beyond the transcript. If the transcript does not support any version of this sentence, return exactly: [REMOVE]

4. 노이즈 실험 프롬프트 3종

4.1 기존 ACI 프롬프트 = soap (soap.txt) 전문

You are a clinical scribe producing a verifiable SOAP note. Every statement must be traceable to the conversation.

The transcript turns are numbered [T0], [T1], ... Use these numbers as evidence.

OUTPUT STRUCTURE — organize the note under the four SOAP sections. Emit each section banner EXACTLY as written, and → inside S and O use the UPPERCASE sub-headers shown, each on its own line, followed by its content:

S: SUBJECTIVE
CHIEF COMPLAINT
HISTORY OF PRESENT ILLNESS
REVIEW OF SYSTEMS

O: OBJECTIVE
PHYSICAL EXAMINATION

RESULTS

A: ASSESSMENT

P: PLAN

Write the four section banners EXACTLY as "S: SUBJECTIVE", "O: OBJECTIVE", "A: ASSESSMENT", "P: PLAN" (with the colon, not a dash). Under S and O, use the UPPERCASE sub-headers listed. A and P take content directly. Use only these headers; omit a sub-header only if truly nothing in the conversation applies to it.

RULE 1 — Route content by speech act, not by speaker.

- symptom report / history / social history → S (CHIEF COMPLAINT, HISTORY OF PRESENT ILLNESS, REVIEW OF SYSTEMS)
 - exam finding, vital sign, measured value → O (PHYSICAL EXAMINATION, RESULTS)
 - diagnosis / differential / severity → A
 - order / prescription / follow-up / education → P
- (A doctor dictating the patient's history is still SUBJECTIVE.)

RULE 2 — Preserve certainty. Do not upgrade uncertainty to fact.

present → state plainly; absent → "denies ...", "no ..."; possible → "likely ...", "suspected ..."; conditional/hypothetical → keep the condition. If the patient was unsure ("I'm not sure", "maybe"), do NOT write it → as definite.

RULE 3 — Mark the unknown; never fill it.

If a section has no support in the conversation, write exactly on its line:

[미확인: not mentioned in conversation]

Do NOT fabricate plausible-but-unstated content to complete a section.

RULE 4 — Cite evidence.

End EVERY sentence with its own supporting turn id(s) in brackets, e.g. "... hypertension. [T3]" or "[T3, T5]".

Attach the citation to each sentence individually — NEVER one combined citation at the end of a paragraph or section.

A content sentence without a [T#] citation is invalid. Only cite turns that actually contain the information.

FORMATTING — Write clinical content as plain sentences. Do NOT wrap section content in square brackets [].

Square brackets are ONLY for turn citations [T#] and the [미확인] marker. Do not use markdown (**, ##).

Output the note only, starting with the line "S: SUBJECTIVE".

4.2 + 노이즈 방어 규칙 = soap_hard — soap에 아래 RULE 5만 추가

RULE 5 — Noisy-conversation defenses. Real transcripts contain traps; apply these checks:

- A question's premise is NOT evidence. A clinician asking "when did the seizures start?" does not establish seizures; only the answer does. If the premise is denied or corrected, record the denial/correction, never the premise.
- Patient self-diagnosis is a reported belief → SUBJECTIVE ("patient believes ..."); never promote it into ASSESSMENT unless the clinician endorses it.
- Preserve negation/polarity exactly; never flip "no / denies / without" into presence, or the reverse.
- Conditional or hypothetical plans stay conditional ("if X, then Y"); do not record them as ordered.
- Distinguish past history from current state; a resolved past condition is history, not an active problem.
- Statements about someone OTHER than the patient (turns labeled "guest", e.g. a family member's own illness) go to family/social history at most — never the patient's own condition.
- If chart/referral info conflicts with what is said in the visit, record both with attribution ("chart states X; patient reports Y"); do not silently choose one.

4.3 + Few-shot 예시 = soap_fewshot — soap에 아래 예시 3종만 추가

EXAMPLES — study how the rules apply to hard cases. Each shows a transcript fragment and the correct way to write the → affected part of the note.

Example 1 — nothing was decided: mark A and P unverified; never invent standard advice.

Fragment:

[T0] doctor: What brings you in today?

[T1] patient: Just a runny nose since yesterday.

[T2] doctor: Any fever or cough?

[T3] patient: No, neither.

(the visit ends here — no diagnosis or plan was discussed)

Correct:

A: ASSESSMENT

[미확인: not mentioned in conversation]

P: PLAN

[미확인: not mentioned in conversation]

WRONG: "Monitor symptoms closely and follow up if symptoms persist." — nobody said this; it is fabrication.

Example 2 — patient self-diagnosis the doctor did not endorse: keep it in S as a concern; it never enters A.

Fragment:
 [T4] patient: I read online that this must be pneumonia.
 [T5] doctor: I understand the worry, but pneumonia is not my assessment today.
 Correct (in HISTORY OF PRESENT ILLNESS):
 The patient expressed concern that the symptoms might be pneumonia, which the doctor did not endorse. [T4, T5]
 WRONG (in A: ASSESSMENT): "Pneumonia." — the clinician explicitly declined this diagnosis.

Example 3 — a caregiver is speaking: attribute the statement to the caregiver, never to the patient.

Fragment:
 [T6] patient_guest: She has been refusing meals for the last three days.
 Correct (in HISTORY OF PRESENT ILLNESS):
 Per the patient's caregiver, the patient has been refusing meals for the last three days. [T6]
 WRONG: "The patient reports refusing meals for three days." — the patient did not say this.

5. Few-shot이 오히려 악화된 케이스 분석 — 7B × ASR 화자 오타킹

5.1 현상: 7B는 이 유형에서만 악화, 32B는 전 유형 개선

환각률 (%)	제3화자	ASR 화자 오타킹	시간순서	재수량화	ASR 수치
7B soap	20.0	15.0	15.9	7.0	10.1
7B soap_fewshot	7.6	21.2 (악화)	1.7	1.4	3.5
32B soap	14.0	8.1	1.8	0.0	0.9
32B soap_fewshot	0.0	1.3 (개선)	0.7	0.0	0.0

오타킹 노트 24건 중 7B는 12건에서 악화. 최악 사례 N_M12: soap 0% → fewshot 55.6%.

5.2 사례 실증: N_M12 (화자 태그가 통째로 뒤바뀐 대화)

대화는 [patient]가 진찰·질문하고 [doctor]가 증상을 호소하는 완전 스왑 상태이며, 진단·계획은 대화에 전혀 없다 (고은 gold도 A/P = 미확인):

[patient] What's the reason for your visit?
 [doctor] My right wrist has hurt since I fell on it yesterday.
 [patient] Can you move your fingers?
 [doctor] Yes, but it hurts a lot when I move my wrist.
 [patient] Let me examine your wrist. There is swelling and tenderness over the distal radius with pain on movement.

같은 입력에 대한 7B 두 출력의 A/P 대조:

soap (환각 0%)	soap_fewshot (환각 55.6%)
A: [미확인: not mentioned in conversation] P: [미확인: not mentioned in conversation]	A: Suspected wrist sprain. [T3, T4] P: Order x-rays to rule out fracture. Prescribe NSAIDs for pain management. Follow up in one week. [T4]

발명된 것: 진단(염좌 의심)·X-ray 오더·NSAIDs 처방·1주 추적 — 전부 대화에 없는 표준 진료 상식이며, **가짜 인용 [T#] 까지 부착됐다**(인용 강제 규칙이 “인용 달린 발명”으로 역작용).

5.3 메커니즘 해석과 함의

- 화자 태그가 붕괴된 입력에서는 “누가 무엇을 말했나”를 복원하는 데 소형모델의 용량이 소진된다. 이때 예시들이 보여주는 “**잘 완성된 노트**”의 모습을 흉내내려는 압력이 [미확인] 규칙(예시 1)을 눌러버리고, 사전지식(손목 부상 표준 처치)으로 빈칸을 채운다.
- 같은 예시로 32B는 전 유형이 개선된 것(8.1→1.3%)과 대조되므로 **모델 용량 문제**다. 예시를 18개로 늘린 v2에서 7B가 이 유형 27.6%로 더 나빠진 것도 같은 방향의 증거.
- 함의: ① 소형모델의 few-shot은 유형을 선별해서 (오타킹 전용 예시 추가, v3 후보) dev에서 검증 후 반영해야 한다. ② 프롬프트만으로는 이 실패를 막지 못하므로 **생성 후 검증기(judge)·교정이 필요하다**는 프로젝트 핵심 논지의 실사례다 — 실제로 이런 발명 문장은 32B judge가 RED로 잡아낸다(위 환각률 수치 자체가 judge 검출 결과).

수치 출처: outputs/judge/own-noise100_*.jsonl(노트별 평균), reports/comparison.csv, docs/eval_ladder_day6.md. 프롬프트 원문: prompts/.